

0.1 Section 5.1 - Proof By Induction

This section really only covers one thing and that proving by induction

Let's start by proving that

$$\sum_{k=1}^n (2k-1) = n^2$$

Proof 1: Break everything up

$$\begin{aligned}\sum_{k=1}^n (2k-1) &= \sum_{k=1}^n (2k) - \sum_{k=1}^n 1 \\ &= 2 \sum_{k=1}^n k - \sum_{k=1}^n 1 \\ &= 2 \cdot \frac{n(n+1)}{2} - n \\ &= n^2 + n - n \\ &= n^2\end{aligned}$$

Proof 2: Represent as a picture (dots)

$$\sum_{k=1}^1 (2k-1) = 1 = 1^2$$

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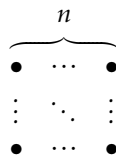
$$\sum_{k=1}^2 (2k-1) = 1 + 3 = 2^2$$

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$$\sum_{k=1}^5 (2k-1) = 1 + 3 + 5 = 1^2$$

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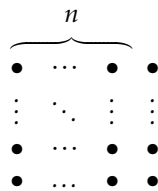
If $\sum_{k=1}^n (2k-1) = n^2$



This is a square matrix so $n \times n$

Then

$$\sum_{k=1}^{n+1} (2k-1) = [1 + 3 + 5 + \dots + (2n-1)] + (2n+1)$$



The extra line is $n \times 1$ and $1 \times n$, so we have $2n$ more dots, and with the extra corner dot, we get

$$n^2 + 2n + 1 = (n+1)^2$$

But we're just assuming that the general formula $= n^2$ is true

Definition 1 (Proof by Induction)

Proof by Induction is basically proving that if a statement about the case n is true, then proving the case $n+1$ will also prove every case

Let $P(n)$ be the proposition function

$$P(n): \sum_{k=1}^n (2k-1) = n^2 \text{ for } n \in \mathbb{Z}^+$$

Base Case: $P(1)$ is true

Inductive Case: $P(n) \implies P(n+1)$

The above implies

$$\begin{aligned} P(1) &\implies P(2) \\ P(2) &\implies P(3) \\ &\vdots \\ P(n) &\implies P(n+1) \end{aligned}$$

So we conclude that $P(n)$ is true $\forall n \in \mathbb{Z}^+$

This is specifically called **Weak Induction**

Strong Induction is

Base Case: $P(1)$ is true

Inductive Case: $P(1) \wedge P(2) \wedge \dots \wedge P(n) \implies P(n+1)$

Let's do some examples

Example 1

Prove the following

1.

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Proof

BASE CASE:

$$\begin{aligned} \sum_{k=1}^1 k^3 &= 1^3 = 1 \\ &= \left[\frac{n(n+1)}{2} \right]^2 = \left[\frac{1(1+1)}{2} \right]^2 = 1 \end{aligned}$$

So the case $n = 1$ is true

INDUCTIVE CASE:

Assume that $\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$

We then prove $\sum_{k=1}^{n+1} k^3 = \left[\frac{(n+1)(n+2)}{2} \right]^2$

Let's do this by making the more complicated part (the proving part) equal to the simpler one

$$\begin{aligned}
\sum_{k=1}^{n+1} k^3 &= \sum_{k=1}^n k^3 + (n+1)^3 = \sum_{k=1}^{n+1} k^3 = \left[\frac{n(n+2)}{2} \right]^2 + (n+1)^3 \\
&= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\
&= (n+1)^2 \cdot \left[\frac{n^2}{4} + (n+1) \right] \\
&= (n+1)^2 \cdot \left[\frac{n^2 + 4n + 4}{4} \right] \\
&= (n+1)^2 \cdot \frac{(n+2)^2}{4} \\
&= \frac{(n+1)^2(n+2)^2}{4} \\
&= \boxed{\left[\frac{(n+1)(n+2)}{2} \right]^2}
\end{aligned}$$

2.

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \text{ for } n \in \mathbb{Z}^+$$

Proof

BASE CASE:

$$\begin{aligned}
\sum_{k=1}^1 k^2 &= 1^2 = 1 \\
\frac{n(n+1)(2n+1)}{6} &= \frac{1 \cdot (1+1)(2+1)}{6} = \frac{2 \cdot 3}{6} = 1
\end{aligned}$$

So our base case $n = 1$ is true

INDUCTIVE CASE:

$$\text{Assume: } \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\text{We prove: } \sum_{k=1}^{n+1} k^2 = \frac{(n+1)(n+2)(2n+3)}{6}$$

Again, let's start with the more complicated looking one and make it look like the simpler one

$$\begin{aligned}
\sum_{k=1}^{n+1} k^2 &= \sum_{k=1}^n k^2 + (n+1)^2 \\
&= \frac{n(n+1)(2n+1)}{6} + \frac{6(n+1)^2}{6} \\
&= (n+1) \left[\frac{n(2n+1)}{6} + \frac{6(n+1)}{6} \right] \\
&= (n+1) \left[\frac{2n^2 + n + 6n + 1}{6} \right] \\
&= (n+1) \left[\frac{2n^2 + 7n + 1}{6} \right] \\
&= (n+1) \left[\frac{(n+2)(2n+3)}{6} \right] \\
&= \boxed{\frac{(n+1)(n+2)(2n+3)}{6}}, \text{ as desired}
\end{aligned}$$

3.

$$2^n \leq n! \text{ for } n \geq 4$$

Proof

BASE CASE:

$$2^4 = 16$$

$$4! = 24$$

$$16 < 24$$

$$\therefore 2^4 \leq 4!$$

So our base case $n = 4$ is true

INDUCTIVE CASE:

Assume: $2^n \leq n!$

We Prove: $2^{n+1} \leq (n+1)!$

$$\begin{aligned}
2^{n+1} &= 2 \cdot 2^n \\
2^{n+1} &\leq 2 \cdot n! \leq (n+1) \cdot n! && (\text{since } 2 \leq (n+1)) \\
&\boxed{= 2^{n+1} \leq (n+1)!}
\end{aligned}$$

Note

This one's kinda lame since we're dealing with inequalities

Let's do another proof but more words

Let's prove:

Every positive interger greater than 1 is a product of prime powers

Proof

BASE CASE: $n = 2$ is a prime power INDUCTIVE CASE:

Assume: n is a product of prime powers

We prove: $n + 1$ is also a product of prime powers

Case 1:

$n + 1$ is a prime

Then $n + 1$ is already a prime power

Case 2:

$n + 1$ is composite

Then $n + 1$ has two factors between 1 and n

1 and n are in our previous list which have prime powers

$\therefore n + 1$ also does

Example 2

Let's do an example with the Fibonacci sequence. Recall that

$$f_n : 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233$$

Let's prove that every 4th Fibonacci number is divisible by 3

In other words,

$$3 \mid f_{4n}, n \in \mathbb{Z}^+$$

Proof

BASE CASE: $n = 1, f_4 = 3, 3 \mid 3$ so the case $n = 1$ is true

INDUCTIVE CASE:

Assume $3 \mid f_{4n}$

We prove $3 \mid f_{4(n+1)}$

$$\begin{aligned} f_{4(n+1)} &= f_{4n+4} = f_{4n+3} + f_{4n+2} \\ &= f_{4n+2} + f_{4n+1} + f_{4n+2} \\ &= 2f_{4n+2} + f_{4n+1} \\ &= 2(f_{4n+1} + f_{4n}) + f_{4n+1} \\ &= 3f_{4n+1} + 2f_{4n} \end{aligned}$$

Since $3 \mid 3f_{4n+1}$ and $3 \mid 2f_{4n}$, then $3 \mid 3f_{4n+1} + 2f_{4n}$

$\therefore 3 \mid f_{4(n+1)}$, as desired